

Remainder theorem



- 1 For each of the following divisors, state the value of x that needs to be substituted into $P(x)$ to find the remainder:

a $x + 3$

b $8 - x$

c $6 - x$

d $x + 10$

e $-7 + 6x$

f $3x - 1$

g $2x + 1$

h $5 + 4x$

- 2 For each of the following, find the remainder when $P(x)$ is divided by $A(x)$:

a $P(x) = -2x^4 + 7x^3 - 3x^2 - 6x - 5, A(x) = x + 2$

b $P(x) = 3x^4 - 3x^3 - 5x^2 + 4x - 5, A(x) = x + 5$

c $P(x) = 3x^4 + 5x^3 - 2x^2 + 6x + 7, A(x) = 3x + 1$

d $P(x) = 2x^4 - 3x^3 + 6x^2 - 10, A(x) = x - 1$

e $P(x) = x^3 - 2x^2 + 9x - 1, A(x) = x - 4$

f $P(x) = 4x^5 - 6x^3 - 7x^2 + 9x, A(x) = 2x - 5$

g $P(x) = 6x^4 - x^3 + 9x^2 + 10x - 8, A(x) = 4x + 2$

h $P(x) = -4x^4 + 6x^3 + 4x^2 - 7x + 7, A(x) = 3x - 1$

- 3 Find the value of k for each of the following:

a The remainder when $3x^3 - 2x^2 - 4x + k$ is divided by $x - 3$ is 47.

b The remainder when $3x^3 + 4x^2 + kx - 3$ is divided by $x + 1$ is -5 .

c The remainder when $3x^3 + 4x^2 + 4x + k$ is divided by $x - 2$ is 52.

d The remainder when $4x^3 - 2x^2 + kx - 1$ is divided by $x - 2$ is 15.

e The remainder when $3x^3 + 2x^2 + x + k$ is divided by $x + 1$ is -3 .

f The remainder when $2x^3 + 2x^2 + kx - 2$ is divided by $x + 3$ is -80 .

g The remainder when $2x^3 - 2x^2 - 4x + k$ is divided by $x + 3$ is -57 .

h The remainder when $4x^3 + 3x^2 + kx - 2$ is divided by $x + 1$ is -147 .

- 4 The polynomials $P(x) = x^3 + 4x^2 - 5x + n$ and $Q(x) = x^3 + 2x + 17$ leave the same remainder when divided by $x + 1$.
Solve for the value of n .

- 5 Consider the polynomial $x^{99} + 1$.

a Find the remainder when the polynomial is divided by $x + 1$.



Factor theorem

- 6 Suppose we are given a polynomial of degree n . What is the name of the process where we find two polynomials of degree less than n that multiply together to give the original?
- 7 What operation is the opposite of factoring?
- 8 When can we say that a polynomial is factored completely?
- 9 Define the factor theorem.
- 10 Show that $x + 2$ is a factor of $P(x) = x^4 + 3x^3 - 7x^2 - 27x - 18$.
- 11 Consider $P(x) = 2x^3 + 8x^2 + x + 4$.
- a Show that $x + 4$ is a factor of the polynomial $P(x)$.
 - b Now, find the other quadratic factor.
- 12 Consider $P(x) = 10 - 17x + 8x^2 - x^3$.
- a Show that $5 - x$ is a factor of the polynomial $P(x)$.
 - b Now, find the other quadratic factor.
- 13 The polynomials $4x^2 - 7x - 15$ and $5x^2 + 13x + k$ have a common factor of $x + p$, where p is an integer.
- a Solve for the value of p .
 - b Solve for the value of k .
- 14 Consider the following polynomials:
- $P(x) = x^4 - 4x^3 - 20x + 3$
 - $Q(x) = x^3 - 4x^2 - 20x + 3$
 - $R(x) = x^4 - 4x^2 - 20x + 3$
 - $S(x) = x^4 - 4x^3 - 20x^2 + 3$

Determine for which polynomial $(x + 3)$ is a factor. Explain your answer.

15 Rewrite the following polynomials as products of linear factors:



a $x^3 + 2x^2 - 13x + 10$

c $x^3 - 9x^2 + 23x - 15$

e $2x^3 - 11x^2 + 19x - 10$

g $2x^3 + x^2 - 7x - 6$

b $x^3 - 6x^2 - x + 30$

d $x^3 + 8x^2 + 17x + 10$

f $5x^3 - 41x^2 + 83x - 15$

h $4x^3 + x^2 - 13x - 10$